In [1]:
```
import numpy as np  #for arrays & mathematical use
import pandas as pd #for working in dataframes
import matplotlib.pyplot as plt #visualizing data
%matplotlib inline
import seaborn as sns #visualizing data
```

In [2]:
```
df = pd.read_csv("Diwali Sales Data.csv", encoding= 'unicode_escape')
```

In [4]:
```
df.shape
```

Out[4]:
```
(11251, 15)
```

In [6]:
```
df.head()
```

Out[6]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	Stat
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtı
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Prades
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Prades
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnatak
4	1000588	Joni	P00057942	M	26-35	28	1	Gujara

Data Cleaning

In [7]:
```
df.info() #gives all the information
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  object 
 2   Product_ID        11251 non-null  object 
 3   Gender            11251 non-null  object 
 4   Age Group         11251 non-null  object 
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  object 
 8   Zone              11251 non-null  object 
 9   Occupation        11251 non-null  object 
 10  Product_Category  11251 non-null  object 
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

In [8]:
```
#drop unrealted/ blank columns
df.drop(['Status','unnamed1'], axis=1, inplace=True)
```

In [9]:
```
pd.isnull(df) #to check the null values
```

Out[9]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Z
0	False	False	False	False	False	False	False	False	F
1	False	False	False	False	False	False	False	False	F
2	False	False	False	False	False	False	False	False	F
3	False	False	False	False	False	False	False	False	F
4	False	False	False	False	False	False	False	False	F
...	...	...	...	...	...	...	...	...	
11246	False	False	False	False	False	False	False	False	F
11247	False	False	False	False	False	False	False	False	F
11248	False	False	False	False	False	False	False	False	F
11249	False	False	False	False	False	False	False	False	F
11250	False	False	False	False	False	False	False	False	F

11251 rows × 13 columns

In [10]:
```
pd.isnull(df).sum()
```

Out[10]:
```
User_ID             0
Cust_name           0
Product_ID          0
Gender              0
Age Group           0
Age                 0
Marital_Status      0
State               0
Zone                0
Occupation          0
Product_Category    0
Orders              0
Amount             12
dtype: int64
```

In [11]:
```
df.shape
```

Out[11]:
```
(11251, 13)
```

In [12]:
```
#to drop the null values
df.dropna(inplace=True)
```

In [13]:
```
df.shape
```

Out[13]: (11239, 13)

In [14]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 11239 entries, 0 to 11250
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11239 non-null  int64
 1   Cust_name         11239 non-null  object
 2   Product_ID        11239 non-null  object
 3   Gender            11239 non-null  object
 4   Age Group         11239 non-null  object
 5   Age               11239 non-null  int64
 6   Marital_Status    11239 non-null  int64
 7   State             11239 non-null  object
 8   Zone              11239 non-null  object
 9   Occupation        11239 non-null  object
 10  Product_Category  11239 non-null  object
 11  Orders            11239 non-null  int64
 12  Amount            11239 non-null  float64
dtypes: float64(1), int64(4), object(8)
memory usage: 1.2+ MB
```

In [15]:
```
#changes in datatyope
df['Amount'] = df['Amount'].astype('int')
```

In [16]:
```
df['Amount'].dtypes
```

Out[16]: dtype('int64')

In [17]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 11239 entries, 0 to 11250
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11239 non-null  int64
 1   Cust_name         11239 non-null  object
 2   Product_ID        11239 non-null  object
 3   Gender            11239 non-null  object
 4   Age Group         11239 non-null  object
 5   Age               11239 non-null  int64
 6   Marital_Status    11239 non-null  int64
 7   State             11239 non-null  object
 8   Zone              11239 non-null  object
 9   Occupation        11239 non-null  object
 10  Product_Category  11239 non-null  object
 11  Orders            11239 non-null  int64
 12  Amount            11239 non-null  int64
dtypes: int64(5), object(8)
memory usage: 1.2+ MB
```

In [18]:
```
df.columns
```

Out[18]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [20]:
```
#to change the name of the column
df.rename(columns= {'Marital_Status':'Shaadi'})
```

Out[20]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Shaadi	State
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat
...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra

11239 rows × 13 columns

In [21]:
```
df.columns
```

Out[21]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [22]:
```
#to get the mean, median, mode and any other things
df.describe()
```

Out[22]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [23]:
```
df[['Age','Orders','Amount']].describe() #to get the specific columns
```

Out[23]:

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Exploratory data analysis (EDA)

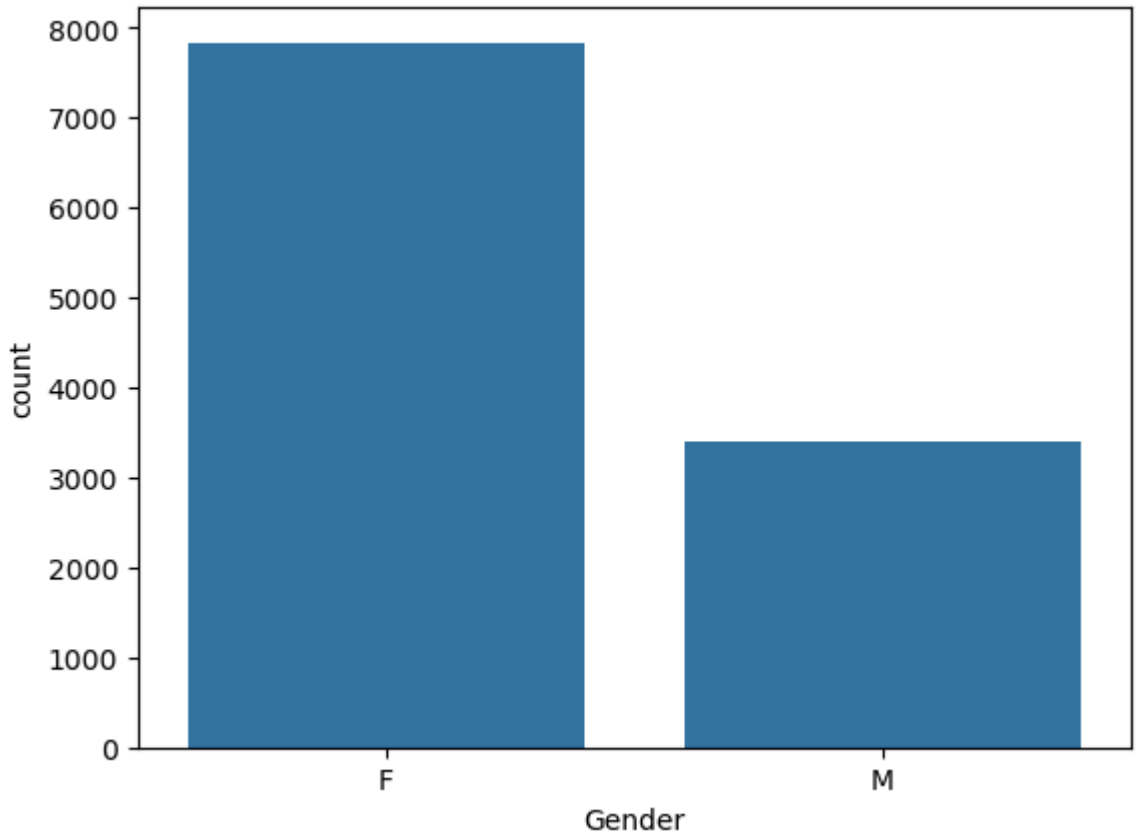
Gender

In [24]:
```
df.columns
```

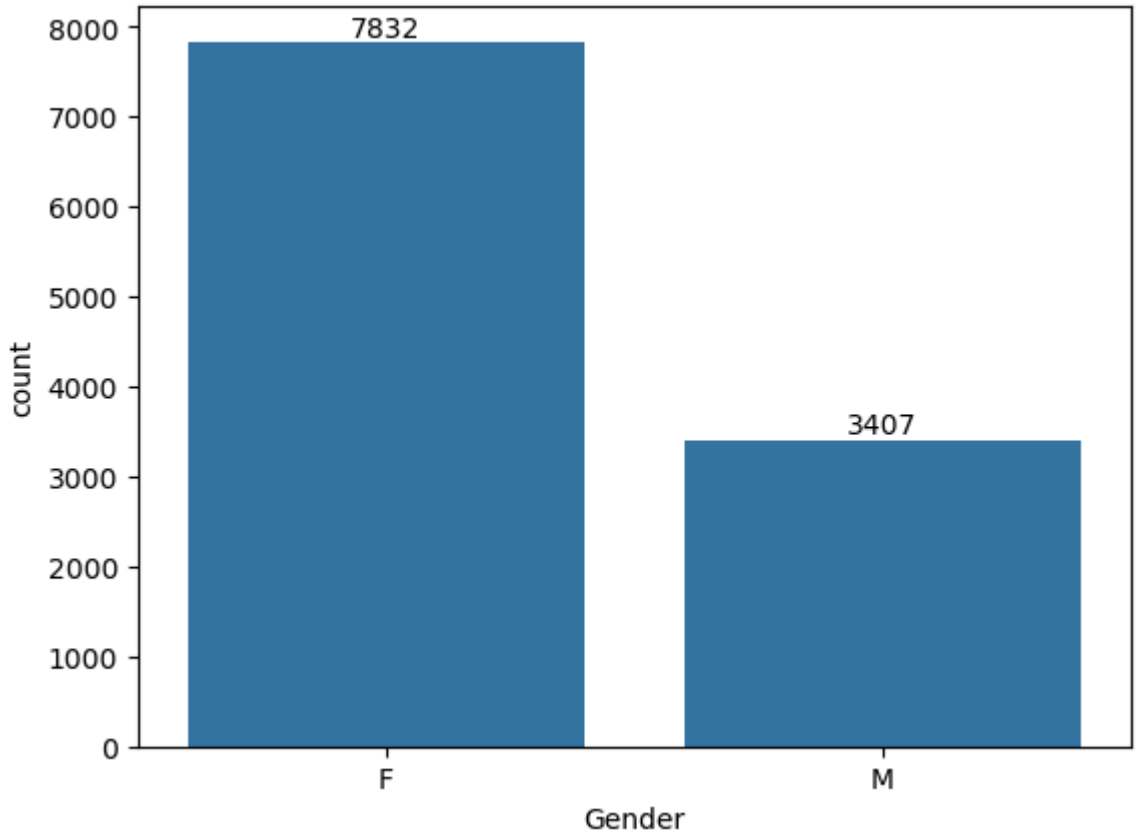
Out[24]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [25]:
```
sns.countplot(x='Gender',data=df)
```

Out[25]:
```
<Axes: xlabel='Gender', ylabel='count'>
```



In [28]:
```
ax = sns.countplot(x = 'Gender',data=df)

for bars in ax.containers:
    ax.bar_label(bars) #to get the labels at the top
```



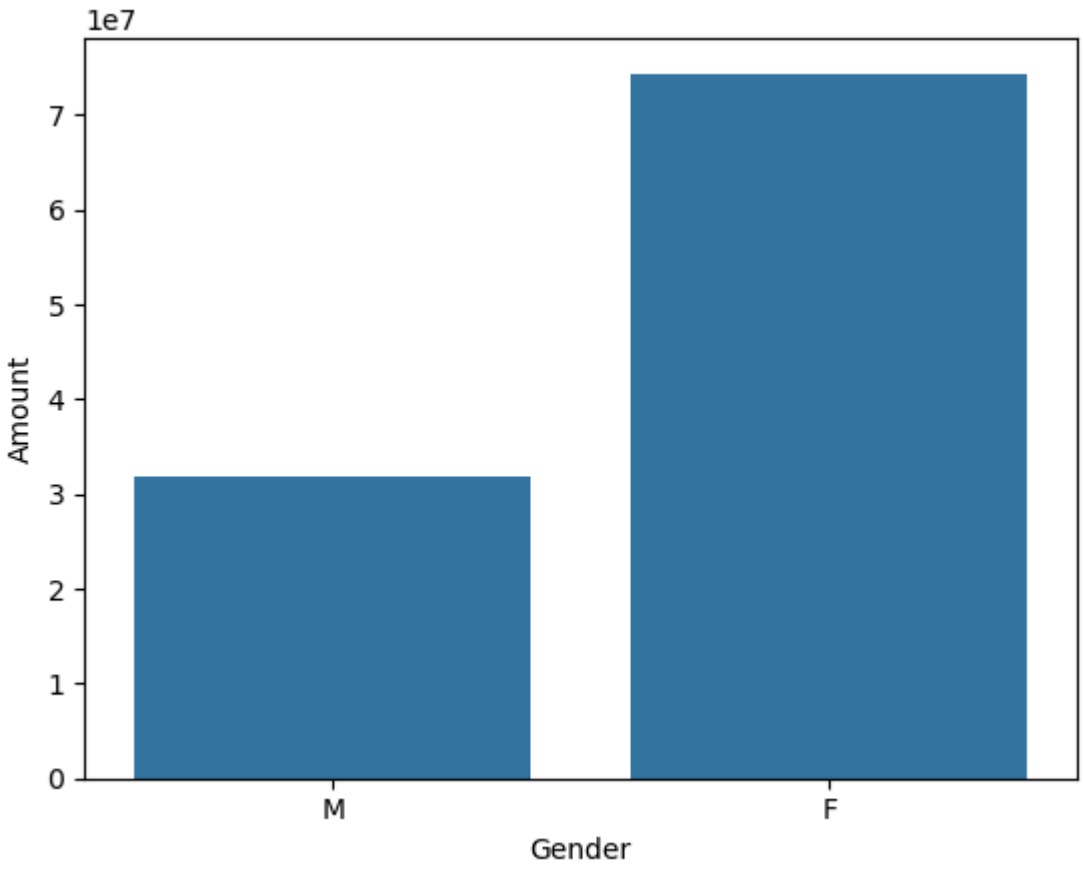
In [30]:
```
df.groupby(['Gender'],as_index=False)['Amount'].sum().sort_values(by='Amount',as
```

Out[30]:

	Gender	Amount
1	M	31913276
0	F	74335853

In [31]:
```
sales_gen = df.groupby(['Gender'],as_index=False)['Amount'].sum().sort_values(by
sns.barplot(x='Gender',y='Amount',data=sales_gen)
```

Out[31]:
```
<Axes: xlabel='Gender', ylabel='Amount'>
```



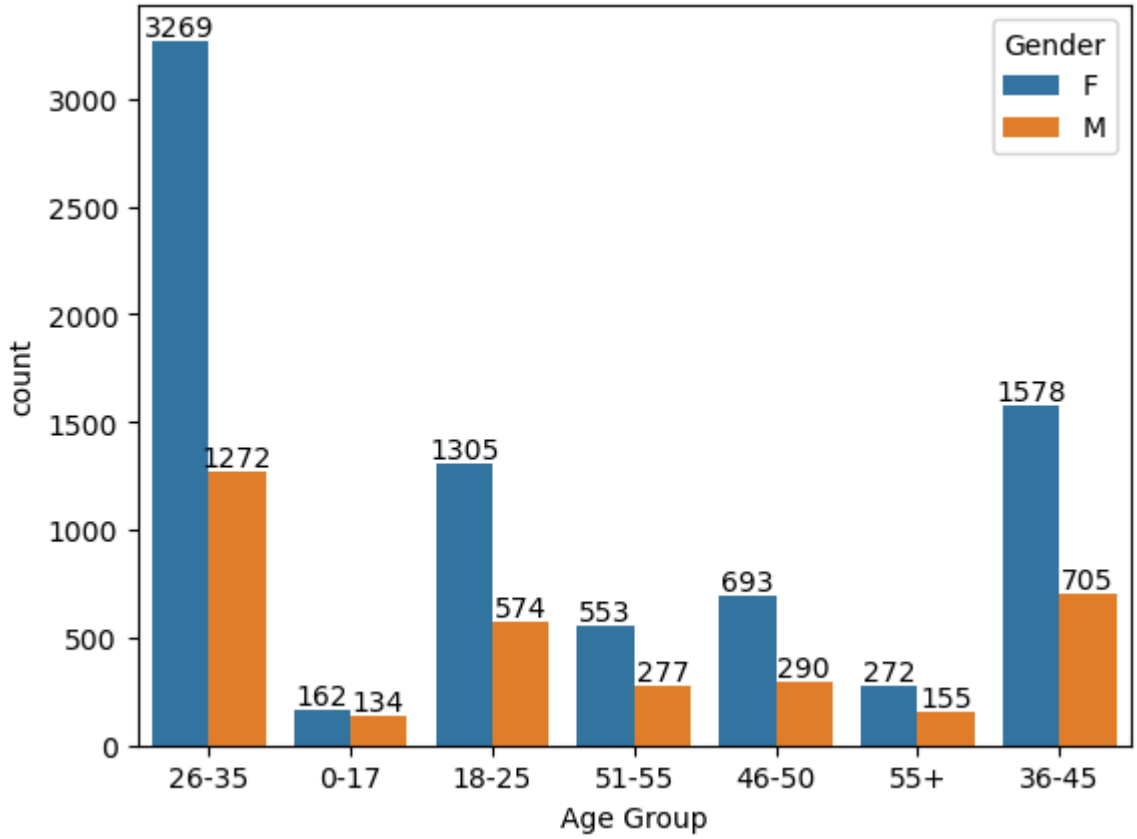
Conclusion = Most of the buyers are female and they do have the puchasing power

Age

In [32]:
```
df.columns
```

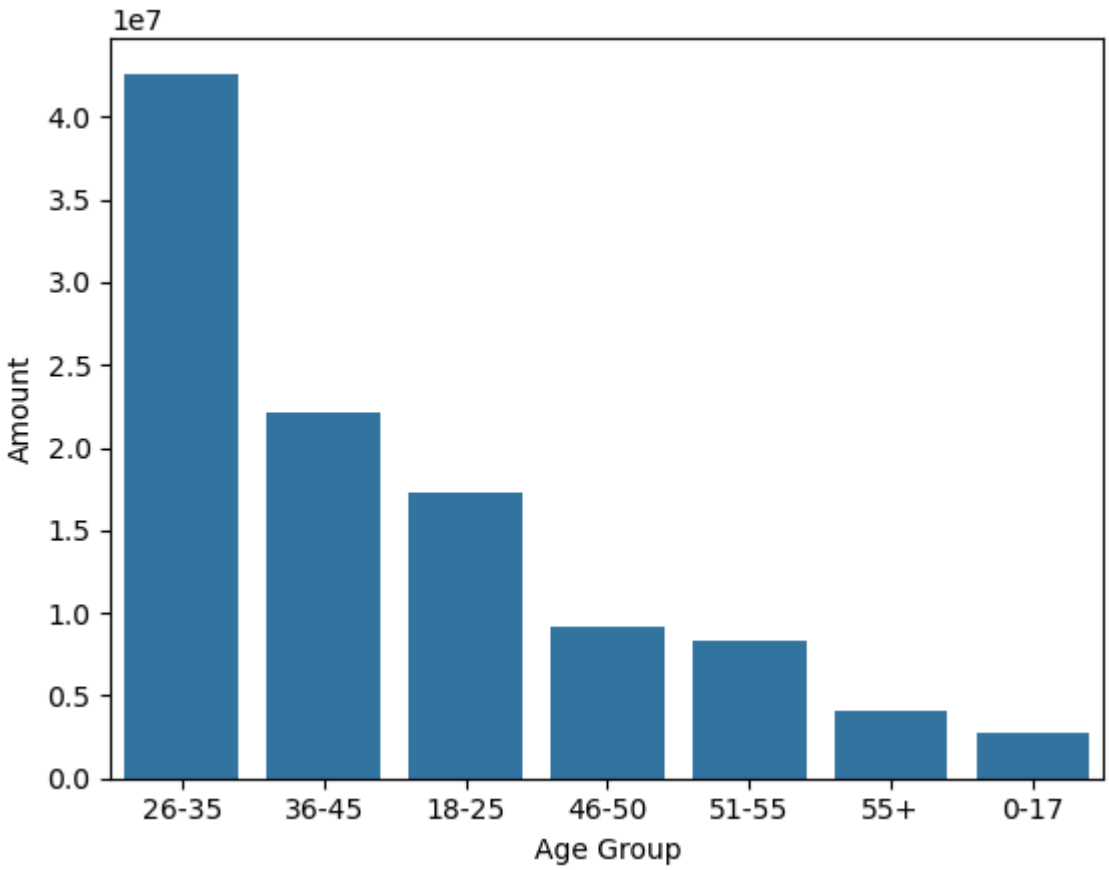
Out[32]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [33]:
```
ax = sns.countplot(data = df,x='Age Group',hue='Gender')
for bars in ax.containers:
    ax.bar_label(bars)
```



In [34]:
```
#Total amount VS Age group
sales_age = df.groupby(['Age Group'],as_index = False)['Amount'].sum().sort_valu
sns.barplot(x='Age Group',y='Amount',data=sales_age)
```

Out[34]: <Axes: xlabel='Age Group', ylabel='Amount'>



Conclusion = Age between 26-35 and mostly female are the buyers

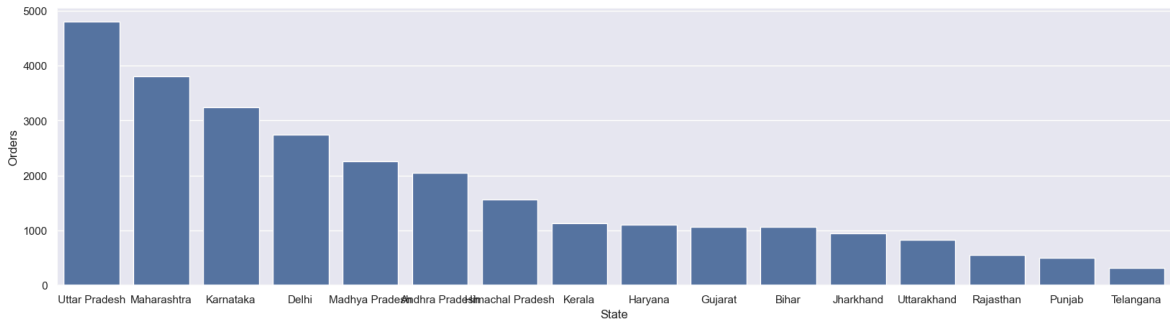
State

In [36]:
```
df.columns
```

Out[36]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

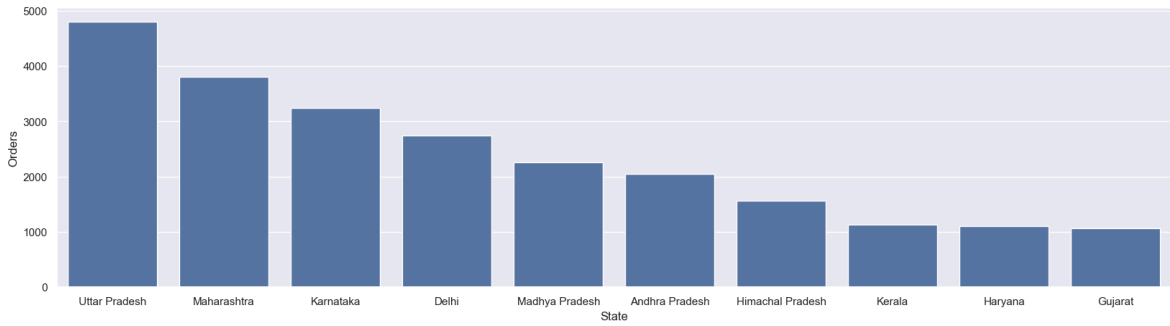
In [44]:
```
#to check the orders state wise
orders_state = df.groupby(['State'],as_index=False)['Orders'].sum().sort_values(
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data=orders_state, x='State', y='Orders')
```

Out[44]:
```
<Axes: xlabel='State', ylabel='Orders'>
```



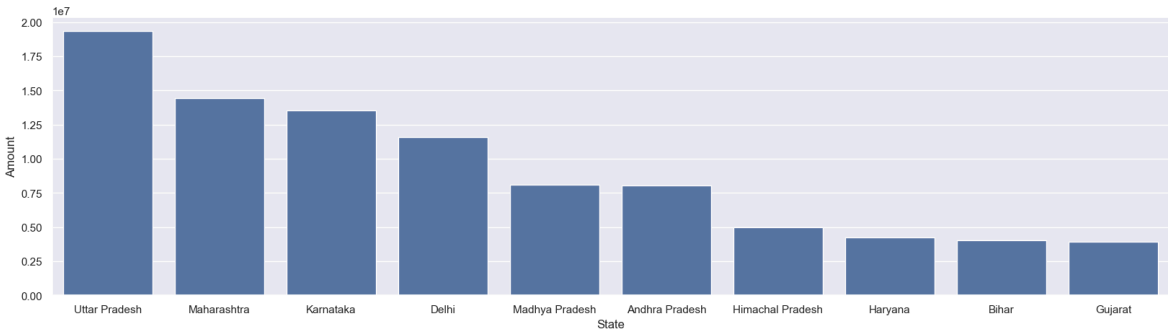
In [45]:
```
#to check the orders state wise top 10
orders_state = df.groupby(['State'],as_index=False)['Orders'].sum().sort_values(
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data=orders_state, x='State', y='Orders')
```

Out[45]:
```
<Axes: xlabel='State', ylabel='Orders'>
```



In [46]:
```
#to check the spending amount state wise top 10
sales_state = df.groupby(['State'],as_index=False)['Amount'].sum().sort_values(b
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data=sales_state, x='State', y='Amount')
```

Out[46]:
```
<Axes: xlabel='State', ylabel='Amount'>
```



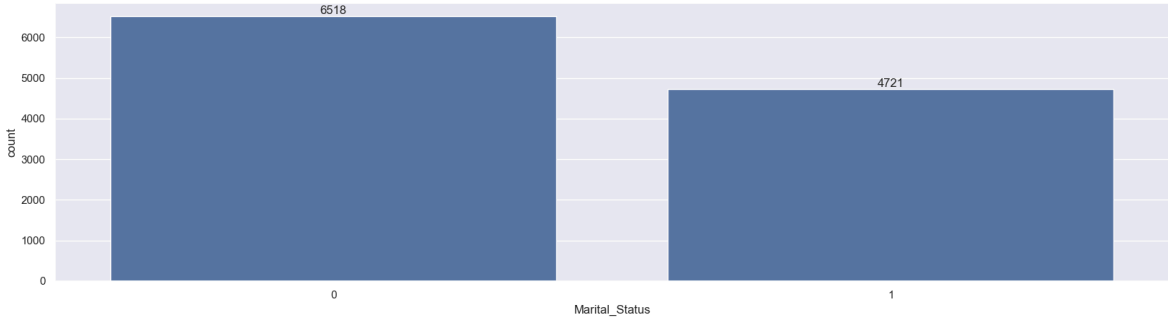
Conclusion = we can see that most of the orders and money spend are from Uttar Pradesh, Maharashtra & Karnataka

Marital Status

In [47]:
```
df.columns
```

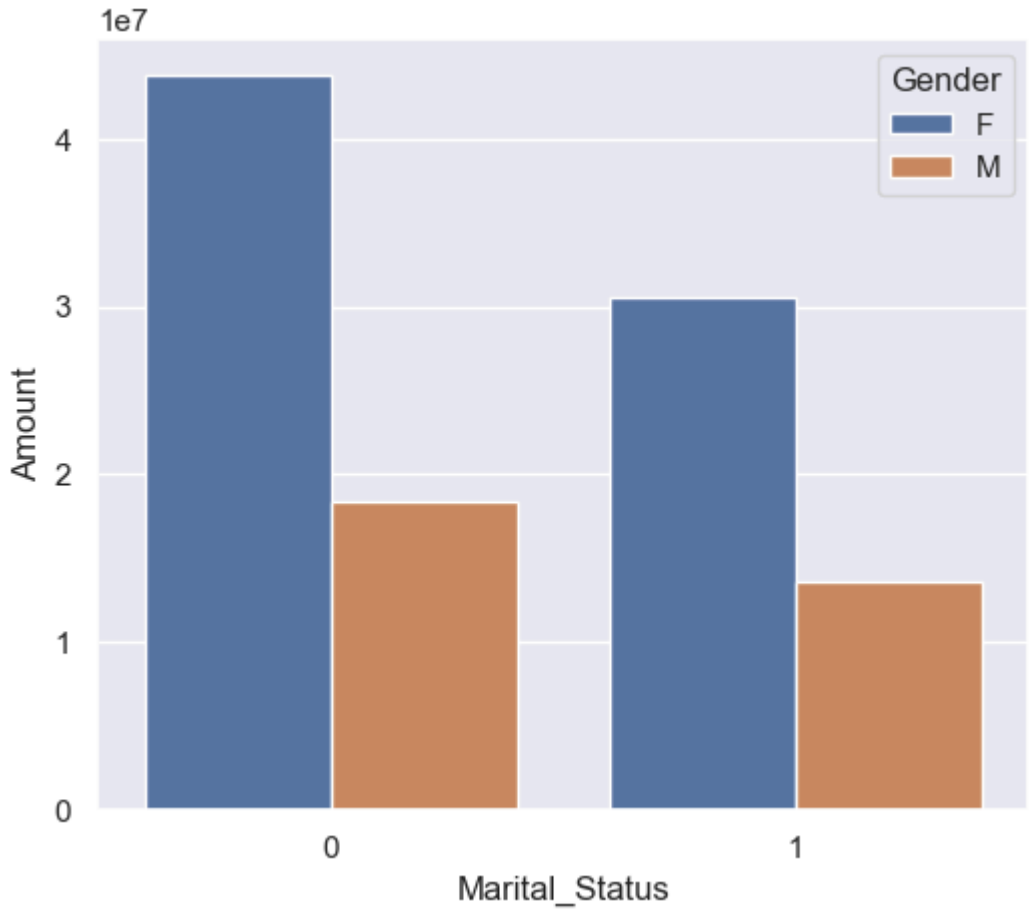
Out[47]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [49]:
```
ax = sns.countplot(data=df,x='Marital_Status')
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



In [53]:
```
sales_data = df.groupby(['Marital_Status','Gender'],as_index=False)['Amount'].su
sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data=sales_data,x='Marital_Status',y='Amount', hue='Gender')
```

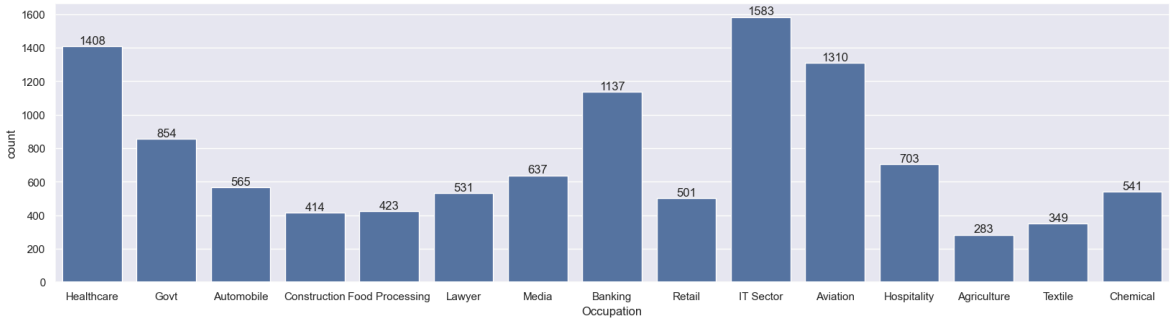
Out[53]:
```
<Axes: xlabel='Marital_Status', ylabel='Amount'>
```



Conclusion = Married women have the most purchasing power

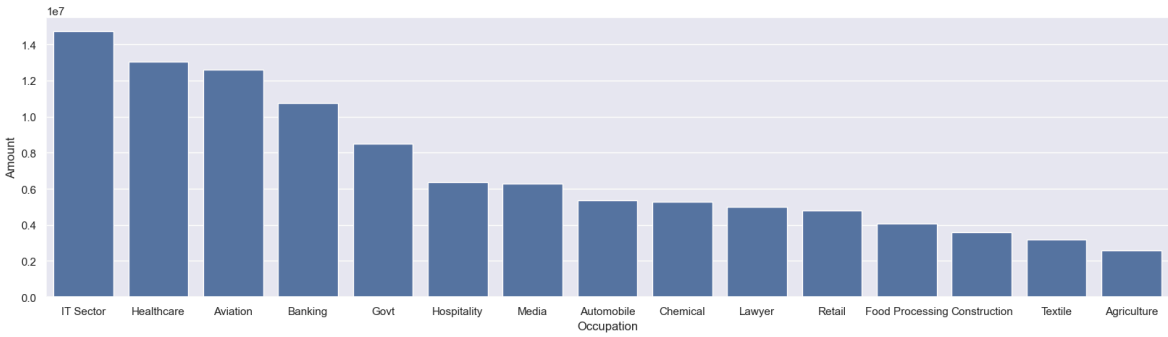
Occupation

In [55]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x='Occupation')
for bars in ax.containers:
    ax.bar_label(bars)
```



In [57]:
```
sales_data = df.groupby(['Occupation'],as_index=False)['Amount'].sum().sort_valu
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data=sales_data,x='Occupation',y='Amount')
```

Out[57]: <Axes: xlabel='Occupation', ylabel='Amount'>



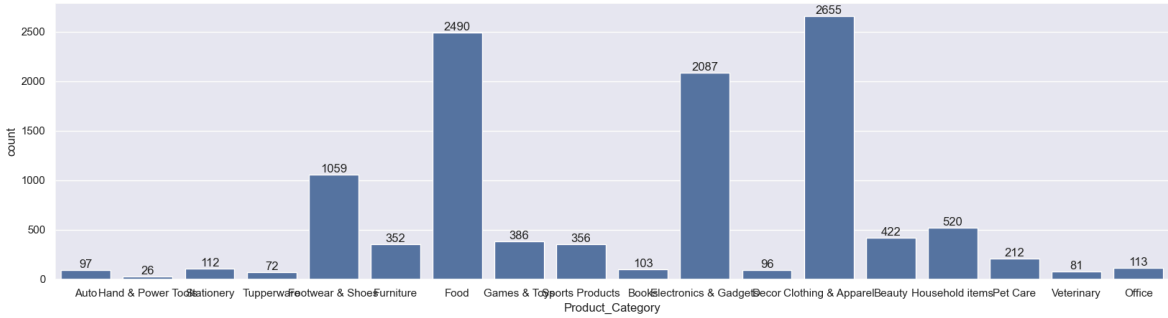
Conclusion = we can see that most buyers are working in IT sector, Healthcare & Aviation Industry

Product Category

In [58]:
```
df.columns
```

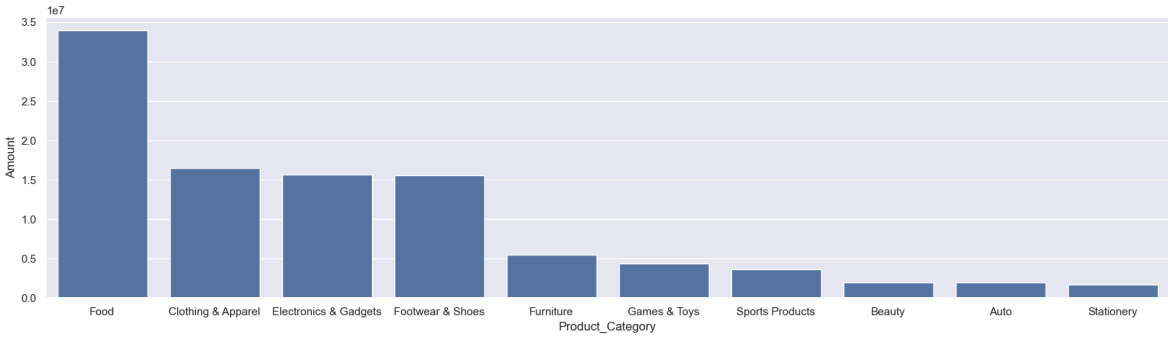
Out[58]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [62]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data=df,x='Product_Category')

for bars in ax.containers:
    ax.bar_label(bars)
```



In [65]:
```
sales_data = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().so

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data=sales_data,x='Product_Category',y='Amount')
```

Out[65]:
```
<Axes: xlabel='Product_Category', ylabel='Amount'>
```

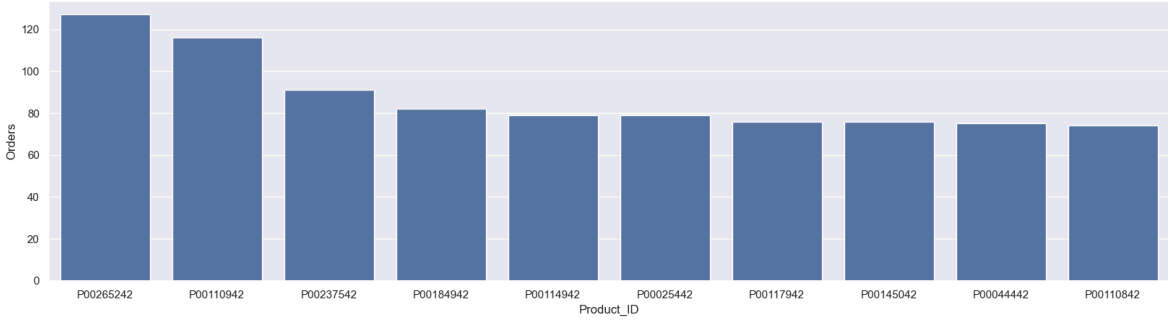


In [67]:
```
df.columns
```

Out[67]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [69]:
```
#which product id has created most sales
sales_data = df.groupby(['Product_ID'],as_index=False)['Orders'].sum().sort_valu
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_data, x='Product_ID',y='Orders')
```

Out[69]:
```
<Axes: xlabel='Product_ID', ylabel='Orders'>
```



Final Conclusion = Married women age group around 26-35 from UP, Maharashtra & Karnataka working in the IT Sector, healthcare & Aviation are more likely to buy products from food, clothes & electronics category.

In [ ]: